

$$\begin{aligned}
& \left(-\frac{2}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_d^{\mathbf{P}}] \delta_{i1i2} + \frac{5}{36} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_d^{\mathbf{P}}] \delta_{i1i2} - \frac{8}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_d^{\mathbf{P}}] \delta_{i1i2} - \right. \\
& \frac{2}{9} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_e^{\mathbf{P}}] \delta_{i1i2} + \frac{5}{12} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_e^{\mathbf{P}}] \delta_{i1i2} - \frac{8}{45} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_e^{\mathbf{P}}] \delta_{i1i2} - \\
& \frac{1}{9} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_l^{\mathbf{P}}] \delta_{i1i2} + \frac{5}{24} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_l^{\mathbf{P}}] \delta_{i1i2} - \frac{4}{45} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_l^{\mathbf{P}}] \delta_{i1i2} - \\
& \frac{1}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_q^{\mathbf{P}}] \delta_{i1i2} + \frac{5}{72} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_q^{\mathbf{P}}] \delta_{i1i2} - \frac{4}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_q^{\mathbf{P}}] \delta_{i1i2} - \\
& \frac{8}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_u^{\mathbf{P}}] \delta_{i1i2} + \frac{5}{9} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_u^{\mathbf{P}}] \delta_{i1i2} - \frac{32}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_u^{\mathbf{P}}] \delta_{i1i2} - \\
& \frac{1}{9} g_1^4 \text{LF}_{3,0} [\tilde{\mu}] \delta_{i1i2} - \frac{1}{6} g_1^4 \text{LF}_{4,-1} [\tilde{\mu}] \delta_{i1i2} + \frac{8}{45} g_1^4 \text{LF}_{5,-2} [\tilde{\mu}] \delta_{i1i2} - \\
& \frac{1}{12} g_1^4 \text{LF}_{2,1,0} [m_1, m_e^{i1}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{2,2,-1} [m_1, m_e^{i1}] \delta_{i1i2} + \\
& \frac{1}{6} g_1^4 \text{LF}_{3,1,-1} [m_1, m_e^{i1}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{4,1,-2} [m_1, m_e^{i1}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{2,1,0} [m_1, m_e^{i2}] \delta_{i1i2} - \\
& \frac{1}{12} g_1^4 \text{LF}_{2,2,-1} [m_1, m_e^{i2}] \delta_{i1i2} + \frac{1}{6} g_1^4 \text{LF}_{3,1,-1} [m_1, m_e^{i2}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{4,1,-2} [m_1, m_e^{i2}] \delta_{i1i2} - \\
& \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \text{LF}_{2,2,0} [m_d^r, m_q^{\mathbf{P}}] \delta_{i1i2} + \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \text{LF}_{3,2,-1} [m_d^r, m_q^{\mathbf{P}}] \delta_{i1i2} - \\
& \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \text{LF}_{2,2,0} [m_e^r, m_l^{\mathbf{P}}] \delta_{i1i2} + \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \text{LF}_{3,2,-1} [m_e^r, m_l^{\mathbf{P}}] \delta_{i1i2} + \\
& \frac{1}{6} g_1^4 \text{LF}_{2,1,0} [m_e^{i1}, m_1] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{3,1,-1} [m_e^{i1}, m_1] \delta_{i1i2} + \frac{1}{6} g_1^4 \text{LF}_{2,1,0} [m_e^{i2}, m_1] \delta_{i1i2} - \\
& \frac{1}{12} g_1^4 \text{LF}_{3,1,-1} [m_e^{i2}, m_1] \delta_{i1i2} + \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \text{LF}_{3,1,0} [m_l^{\mathbf{P}}, m_e^r] \delta_{i1i2} + \\
& \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \text{LF}_{3,2,-1} [m_l^{\mathbf{P}}, m_e^r] \delta_{i1i2} - \frac{3}{4} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \text{LF}_{4,1,-1} [m_l^{\mathbf{P}}, m_e^r] \delta_{i1i2} + \\
& \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \text{LF}_{5,1,-2} [m_l^{\mathbf{P}}, m_e^r] \delta_{i1i2} + \frac{1}{6} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{2,1,0} [m_l^{\mathbf{P}}, \tilde{\mu}] - \\
& \frac{1}{12} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{2,2,-1} [m_l^{\mathbf{P}}, \tilde{\mu}] - \frac{1}{12} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{3,1,-1} [m_l^{\mathbf{P}}, \tilde{\mu}] + \\
& \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \text{LF}_{3,1,0} [m_q^{\mathbf{P}}, m_d^r] \delta_{i1i2} + \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\mathbf{P}}, m_d^r] \delta_{i1i2} - \\
& \frac{5}{4} g_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \text{LF}_{4,1,-1} [m_q^{\mathbf{P}}, m_d^r] \delta_{i1i2} + g_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \text{LF}_{5,1,-2} [m_q^{\mathbf{P}}, m_d^r] \delta_{i1i2} + \\
& \frac{1}{6} g_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} \text{LF}_{2,2,0} [m_q^{\mathbf{P}}, m_u^r] \delta_{i1i2} - \frac{1}{12} g_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\mathbf{P}}, m_u^r] \delta_{i1i2} - \\
& \frac{1}{6} g_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} \text{LF}_{3,1,0} [m_u^r, m_q^{\mathbf{P}}] \delta_{i1i2} - \frac{1}{6} g_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} \text{LF}_{3,2,-1} [m_u^r, m_q^{\mathbf{P}}] \delta_{i1i2} - \\
& \frac{1}{2} g_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} \text{LF}_{4,1,-1} [m_u^r, m_q^{\mathbf{P}}] \delta_{i1i2} + g_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} \text{LF}_{5,1,-2} [m_u^r, m_q^{\mathbf{P}}] \delta_{i1i2} + \\
& \frac{5}{12} g_1^4 \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] \delta_{i1i2} - \frac{1}{3} g_1^4 \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] \delta_{i1i2} + \frac{5}{4} g_1^2 g_2^2 \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] \delta_{i1i2} - \\
& g_1^2 g_2^2 \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] \delta_{i1i2} - \frac{1}{12} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{2,1,0} [\tilde{\mu}, m_l^{\mathbf{P}}] + \\
& \frac{5}{12} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{3,1,-1} [\tilde{\mu}, m_l^{\mathbf{P}}] - \frac{1}{6} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{4,1,-2} [\tilde{\mu}, m_l^{\mathbf{P}}] + \\
& \frac{1}{8} g_1^4 \text{LF}_{2,1,1,-1} [m_1, m_e^{i1}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{4} g_1^4 m_1^2 \text{LF}_{2,1,1,0} [m_1, m_e^{i1}, \tilde{\mu}] \delta_{i1i2} + \\
& \frac{1}{8} g_1^4 \text{LF}_{2,1,1,-1} [m_1, m_e^{i2}, \tilde{\mu}] \delta_{i1i2} + \frac{1}{4} g_1^4 m_1^2 \text{LF}_{2,1,1,0} [m_1, m_e^{i2}, \tilde{\mu}] \delta_{i1i2} + \\
& \frac{1}{8} g_1^4 \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_e^{i1}] \delta_{i1i2} - \frac{1}{8} g_1^4 m_1^2 \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_e^{i1}] \delta_{i1i2} + \\
& \frac{1}{8} g_1^4 \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_e^{i2}] \delta_{i1i2} - \frac{1}{8} g_1^4 m_1^2 \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_e^{i2}] \delta_{i1i2} - \\
& \frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_l^{\mathbf{P}}] - \frac{3}{8} g_2^2 \tilde{\mu}^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_l^{\mathbf{P}}] + \\
& \frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_1, m_l^{\mathbf{P}}] + \frac{3}{4} g_2^$$